

SCOPE: Algorithms for each causal learner

Algorithm 1 SCOPE-S: Training and Inference

Training (Offline)

Input: Dataset $\mathcal{D} = \bigcup_{c=1}^C \bigcup_{k \in \mathcal{K}^{(c)}} \{(\sigma_k^{(c)}, a_k^{obs, (c)}, y^{(c)})\}$, action spaces $\{A_k\}_{k=1}^K$

```

1:  $\hat{V}_{K+1}^{(c)} = y^{(c)} \forall c \in \{1, \dots, C\}; \mathbf{M} = []$  ▷ Initialize
2: for  $k = K, K-1, \dots, 1$  do ▷ Iterate backwards
3:    $\mathcal{C}_k = \{c | k \in \mathcal{K}^{(c)}\}$  ▷ Identify cases that visited  $k$ 
4:    $\mathcal{M}_k \leftarrow \text{Regress } \hat{V}_{k+1}^{(c)} \text{ on } (\sigma_k^{(c)}, a_k^{obs, (c)}) \text{ for all } c \in \mathcal{C}_k$  ▷ S-learner Q-function
5:   Append  $\mathcal{M}_k$  to  $\mathbf{M}$ 
6:   for each case  $c \in \mathcal{C}_k$  do
7:      $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{M}_k(\sigma_k^{(c)}, a_k)$  ▷ Estimated optimal action
8:      $\hat{Q}_k^{obs, (c)} = \mathcal{M}_k(\sigma_k^{(c)}, a_k^{obs, (c)})$  ▷ Q-value of observed action
9:      $\hat{Q}_k^{opt, (c)} = \mathcal{M}_k(\sigma_k^{(c)}, \hat{a}_k^{opt, (c)})$  ▷ Q-value of optimal action
10:     $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)} + \hat{Q}_k^{opt, (c)} - \hat{Q}_k^{obs, (c)}$  ▷ Regret-based value update
11:   end for
12:   for each case  $c \notin \mathcal{C}_k$  do
13:      $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)}$  ▷ Carry over value if point  $k$  is skipped
14:   end for
15: end for
16: return  $\mathbf{M}$ 

```

Inference (Online)

Input: Models $\{\mathcal{M}_k\}_{k=1}^K$, initial prefix $\sigma_{k_{first}}^{(c)}$

```

17: while case  $c$  is not completed do
18:   if case  $c$  reaches DECISION POINT  $k \in \{1, \dots, K\}$  then
19:     Observe current prefix  $\sigma_k^{(c)}$ 
20:      $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{M}_k(\sigma_k^{(c)}, a_k)$ 
21:     Execute action  $\hat{a}_k^{opt, (c)}$ 
22:   end if
23: end while

```

Algorithm 2 SCOPE-T: Training and Inference

Training (Offline)

Input: Dataset $\mathcal{D} = \bigcup_{c=1}^C \bigcup_{k \in \mathcal{K}^{(c)}} \{(\sigma_k^{(c)}, a_k^{obs, (c)}, y^{(c)})\}$, action spaces $\{A_k\}_{k=1}^K$

```

1:  $\hat{V}_{K+1}^{(c)} = y^{(c)} \forall c \in \{1, \dots, C\}$ ;  $\mathbf{M} = []$  ▷ Initialize
2: for  $k = K, K-1, \dots, 1$  do ▷ Iterate backwards
3:    $\mathcal{C}_k = \{c \mid k \in \mathcal{K}^{(c)}\}$  ▷ Identify cases that visited  $k$ 
4:    $\mathcal{M}_k = []$  ▷ Fit a separate model for each action
5:   for each action  $a_k \in A_k$  do
6:      $\mathcal{M}_k[a_k] \leftarrow \text{Regress } \hat{V}_{k+1}^{(c)} \text{ on } \sigma_k^{(c)} \text{ for } c \in \mathcal{C}_k \text{ s.t. } a_k^{obs, (c)} = a_k$  ▷ T-learner Q-function
7:   end for
8:   Append  $\mathcal{M}_k$  to  $\mathbf{M}$ 
9:   for each case  $c \in \mathcal{C}_k$  do
10:     $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{M}_k[a_k](\sigma_k^{(c)})$  ▷ Estimated optimal action
11:     $\hat{Q}_k^{obs, (c)} = \mathcal{M}_k[a_k^{obs, (c)}](\sigma_k^{(c)})$  ▷ Q-value of observed action
12:     $\hat{Q}_k^{opt, (c)} = \mathcal{M}_k[\hat{a}_k^{opt, (c)}](\sigma_k^{(c)})$  ▷ Q-value of optimal action
13:     $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)} + \hat{Q}_k^{opt, (c)} - \hat{Q}_k^{obs, (c)}$  ▷ Regret-based value update
14:   end for
15:   for each case  $c \notin \mathcal{C}_k$  do
16:      $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)}$  ▷ Carry over value if point  $k$  is skipped
17:   end for
18: end for
19: return  $\mathbf{M}$ 

```

Inference (Online)

Input: Model lists $\{\mathcal{M}_k\}_{k=1}^K$, initial prefix $\sigma_{k_{first}}^{(c)}$

```

20: while case  $c$  is not completed do
21:   if case  $c$  reaches DECISION POINT  $k \in \{1, \dots, K\}$  then
22:     Observe current prefix  $\sigma_k^{(c)}$ 
23:      $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{M}_k[a_k](\sigma_k^{(c)})$ 
24:     Execute action  $\hat{a}_k^{opt, (c)}$ 
25:   end if
26: end while

```

Algorithm 3 SCOPE-RA: Training and Inference

Training (Offline)

Input: Dataset $\mathcal{D} = \bigcup_{c=1}^C \bigcup_{k \in \mathcal{K}^{(c)}} \{(\sigma_k^{(c)}, a_k^{obs, (c)}, y^{(c)})\}$, action spaces $\{A_k\}_{k=1}^K$

```

1:  $\hat{V}_{K+1}^{(c)} = y^{(c)} \forall c \in \{1, \dots, C\}; \mathbf{M} = [ ]; \mathbf{E} = [ ]$  ▷ Initialize
2: for  $k = K, K-1, \dots, 1$  do ▷ Iterate backwards
3:    $\mathcal{C}_k = \{c | k \in \mathcal{K}^{(c)}\}$  ▷ Identify cases that visited  $k$ 
4:    $\mathcal{M}_k \leftarrow \text{Regress } \hat{V}_{k+1}^{(c)} \text{ on } (\sigma_k^{(c)}, a_k^{obs, (c)}) \text{ for all } c \in \mathcal{C}_k$  ▷ S-learner to estimate Q-function
5:   Append  $\mathcal{M}_k$  to  $\mathbf{M}$ 
6:   for each case  $c \in \mathcal{C}_k$  do
7:     for each action  $a_k \in A_k$  do
8:        $\Phi_k^{(a_k)(c)} \leftarrow \text{Compute pseudo-outcome using } (\sigma_k^{(c)}, \mathcal{M}_k, a_k)$ 
9:     end for
10:  end for
11:   $\mathcal{E}_k \leftarrow \text{Regress } \Phi_k^{(a_k)(c)} \text{ on } (\sigma_k^{(c)}, a_k) \text{ for all } c \in \mathcal{C}_k$  ▷ Model to estimate causal effects
12:  Append  $\mathcal{E}_k$  to  $\mathbf{E}$ 
13:  for each case  $c \in \mathcal{C}_k$  do
14:     $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{E}_k(\sigma_k^{(c)}, a_k)$  ▷ Estimated optimal action
15:     $\hat{Q}_k^{obs, (c)} = \mathcal{M}_k(\sigma_k^{(c)}, a_k^{obs, (c)})$  ▷ Q-value of observed action
16:     $\hat{Q}_k^{opt, (c)} = \mathcal{M}_k(\sigma_k^{(c)}, \hat{a}_k^{opt, (c)})$  ▷ Q-value of optimal action
17:     $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)} + \hat{Q}_k^{opt, (c)} - \hat{Q}_k^{obs, (c)}$  ▷ Regret-based value update
18:  end for
19:  for each case  $c \notin \mathcal{C}_k$  do
20:     $\hat{V}_k^{(c)} = \hat{V}_{k+1}^{(c)}$  ▷ Carry over value if point  $k$  is skipped
21:  end for
22: end for
23: return  $\mathbf{E}$  (and  $\mathbf{M}$ )

```

Inference (Online)

Input: Causal effect models $\{\mathcal{E}_k\}_{k=1}^K$, initial prefix $\sigma_{k_{first}}^{(c)}$

```

24: while case  $c$  is not completed do
25:   if case  $c$  reaches DECISION POINT  $k \in \{1, \dots, K\}$  then
26:     Observe current prefix  $\sigma_k^{(c)}$ 
27:      $\hat{a}_k^{opt, (c)} = \text{argmax}_{a_k \in A_k} \mathcal{E}_k(\sigma_k^{(c)}, a_k)$ 
28:     Execute action  $\hat{a}_k^{opt, (c)}$ 
29:   end if
30: end while

```
